

Reflecting on Sustainability, Ethics, and Risk in the Interactive Sticker Dispenser Project

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Abstract

This report reflects on the development of the Interactive Sticker Dispenser for NMES outreach events aimed at GCSE-age students. It replaced the previous process of manually handing out stickers using an interactive electromechanical system. For the device to function effectively as an outreach tool, it needed to be portable, reliable, safe, visually engaging, and maintainable.

I was responsible for the chassis of the dispenser. Using Fusion 360, I designed the enclosure, fabricated the structural components using a laser cutter, and assembled the chassis. I also added a slide-out drawer to house the screen, redesigned battery access and tape-roll replacement at the rear of the unit, and added windows so users could view the operation of the internal mechanisms without compromising safety.

The report covers four themes from the module: Project Management and Stakeholder Dialogue, Sustainability and Ethics, Life Cycle Assessment (LCA), and Failure Modes and Effects Analysis (FMEA). The central argument is that the project required resolving real engineering trade-offs involving educational value, reliability, manufacturability, maintainability, and environmental responsibility. I conducted a simplified LCA comparison of three different materials to consider as options for building the chassis and used a simplified FMEA to assess three aspects related to the chassis: the screen drawer, the battery-access arrangement, and the viewing and chute assembly. The report shows that even a small public prototype required systems thinking to evaluate how every decision regarding design impacted safety, maintainability, user experience, and ultimately environmental impact.

1 Introduction

The Interactive Sticker Dispenser had a simple public function, but the engineering task behind it was more complex. The aim for this project was to turn the manual method of giving out stickers into an engaging and interactive experience with a robot dispenser.

My contribution to the project focused on the chassis of the sticker dispenser. My tasks were to model the design, fabricate the parts, assemble all components including screen integration, provide rear maintenance access, and develop an aesthetically coherent appearance for the device. They defined both portability and usability, and ultimately provided the framework for maintaining the device and presenting it to the general public.

2 Project Management, Values Thinking, and Stakeholder Dialogue

Project management became important once the project moved from design into fabrication and integration. One of the clearest lessons was that a complete CAD model did not guarantee

a manufacturable result. During laser cutting, some parts were initially modelled to the full width of the plywood sheet, which was 300 mm. In practice, the laser cutter was not perfectly aligned with the stock, so the first cut removed material from the sheet edge and the part had to be remade. This showed that fabrication depends on machine tolerances, setup accuracy, and design margin, not only on nominal dimensions.

A second challenge emerged during integration. The battery did not initially fit within the rear structure, so I redesigned the rear bar to create the required clearance. The screen position also created a problem. Although a side-mounted screen appeared feasible in CAD, it would have made transport harder and increased the chance of damage. To solve this, I designed a pull-out drawer with a positive stop so the screen could stay protected during transport and move out during use.

At the proposal stage, the chassis appeared lower risk than the cutting and feed subsystems. However, fabrication tolerances and integration constraints made the chassis highly iterative. Consequently, the enclosure evolved from a straightforward packaging task into a critical subsystem affecting transport, maintenance, and final integration. To manage this uncertainty, planning tools like our Kanban board and sprint-style tasking proved invaluable.

Values thinking played a central role in shaping the enclosure design. Our approach to resolving design conflicts was informed by the *Dilemma* board game played during module lectures, which highlighted how opposing values must be negotiated. For our chassis, the main conflict was between transparency and robustness. From an outreach perspective, visibility was important because younger users are more likely to engage with a system when they can see the mechanism move. At the same time, increasing visibility required more acrylic, which reduced structural resilience during transport. The design therefore had to carefully balance educational value against durability.

Stakeholder dialogue shaped several final decisions. NMES needed a device that was portable, safe, and easy to maintain across repeated events, while younger users responded more strongly to novelty, visual appeal, and intuitive interaction. A key strategy was to translate stakeholder disagreement into explicit design criteria, enabling structured trade-off evaluation. Screen position, viewing panel coverage, and rear access were discussed directly against transport, safety, maintenance, and educational visibility. The final chassis therefore used a compact Wall-E-inspired form, three acrylic viewing panels, and hinged rear access.

3 Sustainability and Ethics in Engineering

Sustainability was not simply about making a sustainable material choice; it was a systems question. The first lecture exercise asked us to select a single SDG and illustrate its relationships with other SDGs through three types of relationship: synergies, anti-synergies and both enablers and obstacles. After centring the mapping of relationships about SDG 4 Quality Education, we found that SDG 9 Industry, Innovation and Infrastructure formed a synergy. We saw this synergy because there were multiple ways in which developing engineering prototypes could enhance the quality of education while they are being used as test beds for new forms of infrastructure. We also identified SDG 12 Responsible Consumption and Production as an anti-synergy, because, by their very nature, the consumption of resources is required when using

prototype-based methods for teaching students. Finally, we identified SDG 17 Partnerships for Goals as an enabling factor. This is based upon our belief that the partnerships established between King's College London and local schools will increase the range of institutions that can be reached during outreach activities. However, SDG 8 Decent Work and Economic Growth may also act as an obstacle if commercialisation pressures shift outreach priorities away from educational purposes and towards promotion.

We identified that the strongest positive relationship was with SDG 4 – Quality Education. The dispenser transformed a passive hand-out into an active engineering experience. As demonstrated in research on STEM outreach, key drivers for student participation include participant visibility, agency, and active participation [1]. All these areas were addressed through our design. The transparent mechanisms, screen input, audio feedback, and animated "eyes" increased agency and provided users with visual representations of cause and effect. Although the device did not teach formal equations, the causal relationships between inputs and outputs were easily understood.

We also recognised a positive relationship with SDG 9, as the device served as a public-facing engineering testbed requiring subsystem integration and iterative testing.

Conversely, our analysis highlighted a negative impact on SDG 12. The device uses wood, plastic, paint, mechanical fasteners, and batteries, and the iterative prototyping process generated scrap waste. The educational value provided by the project inherently consumes resources, highlighting a genuine contradiction. The sustainability of the project, therefore, depends heavily on its continued reuse across multiple events to offset its initial fabrication burden. An open-to-the-public demonstrator has the potential to normalise resource-heavy engineering unless the designer actively incorporates reuse and maintenance plans.

The third lecture exercise focused on formulating potential ethical issues as moral dilemmas. For our project, we had to weigh the educational benefit of open mechanisms against user safety. Users need to observe the operating mechanism to understand the system, but they must also be protected from moving parts. To resolve this dilemma, I incorporated strategic acrylic viewing panes and a narrow output chute, allowing users to safely visually follow the path of their stickers while preventing physical access to pinch points.

The camera-based tracking feature introduces privacy concerns related to data use and user awareness. From an engineering ethics perspective, the use of this feature is justified provided that it complies with established data privacy standards and ensures informed user awareness. If processing is performed locally and users are informed that no personally identifiable information is stored, then it is acceptable to use this technology under adult supervision. This stance is consistent with the ethical principles set out by the Engineering Council and the Royal Academy of Engineering on ethical standards for engineers [2]. The ethical case for the project therefore depends on repeated reuse across multiple events rather than on one successful demonstration.

4 Simplified Life Cycle Assessment

A simplified LCA is suitable here as a comparative design tool [3] since the primary design question revolves around material choice over the life cycle of the product (manufacture, transport, use, disposal). For this project, the enclosure was the primary subsystem where material

selection directly impacted weight, repairability, durability, and educational visibility.

4.1 Goal and Scope

The functional unit is one chassis assembly that holds the sticker dispenser for at least one full year of repeated outreach activities. The assessment boundaries are based on a cradle-to-grave analysis. The assessment includes material production, fabrication, finishing processes, transportation between events, maintenance, and end-of-life scenarios.

Electronics were excluded because they do not contribute to the comparison of the enclosure alternatives themselves.

Three material strategies were considered. Option A used a plywood-dominant chassis with acrylic viewing panels. Option B used an MDF-dominant chassis with acrylic viewing panels. Option C used an aluminium frame with acrylic viewing panels.

4.2 Inventory and Impact Assessment

A spreadsheet-based simplified LCA was conducted for the three chassis options. The chassis wood mass was estimated from CAD part areas and material thicknesses, while acrylic mass was estimated from the viewing panel geometry. Simplified lookup values and proxy values were then used to compare manufacturing, transport, and end-of-life burdens across the three alternatives.

Table 1: Spreadsheet-based simplified LCA impact scores for chassis material options

Option			Manufacturing (kg CO ₂ eq)	Transport (kg CO ₂ eq)	End-of-life (kg CO ₂ eq)	Total (kg CO ₂ eq)
A:	Plywood	+	2572.062	0.024	101.892	2673.978
	acrylic					
B:	MDF	+	3887.077	0.029	137.661	4024.767
	acrylic					
C:	Aluminium	+	38340.468	0.098	514.281	38854.846
	acrylic					

Option A represents the most suitable solution based on a balance of structural performance, manufacturability, and repairability. The plywood supplied an appropriate amount of structural capacity and a favourable ratio of stiffness to weight. Additionally, the plywood was compatible with laser cutting technology and easily permitted fast recutting or rapid replacement when necessary due to the need for iteration in the development process. These factors were particularly relevant in a student project where access to specialised manufacturing technologies and the time available to revise designs were both limited.

4.3 Interpretation

The simplified LCA indicates that Option A, the plywood-dominant chassis with acrylic viewing panels, is the preferred alternative. As shown in Table 1, its total estimated impact score was 2673.978, compared with 4024.767 for Option B and 38854.846 for Option C. This means that

the MDF-based option produced a higher overall burden than plywood, while the aluminium-based option was substantially higher than both.

Table 2: Engineering interpretation of the three chassis alternatives

Criterion	A: Plywood + acrylic	B: MDF + acrylic	C: Aluminium + acrylic
Material production	Moderate	Moderate-high	High
Fabrication burden	Low-moderate	Low	Moderate-high
Portability impact	Good	Poorer	Good
Durability in outreach use	Good	Moderate	Very good
Ease of repair and remake	Good	Good	Moderate
Educational visibility	Good	Good	Good
End-of-life quality	Moderate	Poorer	Moderate
Overall judgement	Best balance	Cost-led but weaker	Durable but unjustified

This result is consistent with the practical experience of fabrication and integration during development. Plywood provided adequate stiffness and strength for the chassis while remaining comparatively lightweight, easy to laser cut, and straightforward to remake when design revisions were required. These qualities were particularly important in a student project context, where workshop access, fabrication time, and the need for rapid iteration were major constraints. By contrast, MDF would increase the overall impact score while also reducing portability and making the structure more vulnerable to handling damage. Aluminium would likely provide the greatest long-term durability, but its much higher impact score and greater fabrication complexity make it difficult to justify for a prototype of this type.

The results also clarify the role of acrylic in the final design. Acrylic viewing panels increase the environmental burden of the chassis compared with an entirely wooden enclosure, but they also contribute directly to the educational purpose of the device by allowing users to observe the internal mechanism safely. It also supports the outreach value of the system. The most defensible design approach is therefore to use acrylic selectively only where visibility meaningfully improves learning, engagement, and safe observation. Similarly, paint and other coatings should be minimised. While they may improve appearance and reinforce the product identity, they can also reduce material purity and complicate end-of-life recovery.

However, the results must be interpreted as a simplified comparative assessment rather than an absolute environmental account of the entire sticker dispenser. The LCA was intentionally limited to the chassis subsystem because the aim was to compare material choices for the enclosure rather than to evaluate electronics, motors, control hardware, or sensing components

that were common to the wider system. The calculations also rely on simplified lookup values and proxy assumptions for some materials, and they do not include supplier-specific data, exact fabrication energy, or detailed transport distances for every life-cycle stage. For this reason, the values should be used mainly to compare alternatives relative to one another rather than as precise environmental totals.

Overall, the simplified LCA supports the conclusion that plywood was the most appropriate chassis material for this project. It achieved the lowest total estimated impact score of the three options while also supporting ease of manufacture, repair, and repeated reuse. The environmental case for the final design therefore depends not only on material selection, but also on continued reuse of the dispenser across multiple NMES outreach events. Under those conditions, the plywood-and-acrylic chassis can be justified as the most balanced solution from both an engineering and sustainability perspective.

5 Simplified Failure Modes and Effects Analysis

I applied a simplified FMEA to three chassis-related details: the screen drawer, the access for the batteries, and the viewing and chute assembly [4]. These have been chosen as they relate to some of the most important aspects of usability, maintainability, safety, and functionality that users expect from a public use item. The screen drawer affects both transportability and user interaction. The battery-access affects maintenance and also is related to electrical packaging. The viewing and chute affect the overall success of the dispensing process.

Table 3: Simplified FMEA for three chassis-related details

Item	Failure mode	S	O	D	RPN	Recommended action
Screen drawer	Drawer jams, overextends, or creates pinch risk	3	3	3	27	Add positive stop. Improve runner alignment. Verify travel under repeated use.
Rear battery access	Battery shifts, cables strain, or replacement is awkward	4	2	4	32	Add restraint and cable management. Define replacement procedure. Inspect before events.
Viewing and chute assembly	Sticker exits poorly or acrylic and chute geometry causes obstruction or cracking	3	3	4	36	Refine chute geometry. Reduce high-contact edges. Allow quick maintenance access.

The viewing and chute assembly was assigned the highest RPN, indicating it represents the most critical reliability risk in the system. Although it created no greater physical hazard than other components, if the sticker does not leave the device properly, users will believe that the entire interaction has been unsuccessful. To address this problem, I used the lower part of the chute for electronic protection and for directing the sticker down a much straighter path. The detection score was relatively high due to the fact that there are few signs that an impending

jam exists and when failure does occur during normal usage, it is not always easy to tell where it occurred.

The battery-access arrangement received the highest severity rating since it represented both servicing needs and electrical packaging. Initially, the battery could not be installed at the back of the device due to insufficient space. Therefore, I redesigned the rear bar to provide adequate room. This experience illustrates the importance of considering serviceability issues early in the process of developing products. For all of these reasons, opening up the rear of the device via a hinge was a required engineering solution and not simply a convenience.

The screen drawer solved portability and usability problems while introducing several new potential hazards including jammed conditions, over-travelling conditions and pinch point injuries. An integrated stop was implemented to mitigate over-travel and pinch risks, directly reducing both severity and occurrence ratings.

6 Conclusion

This project demonstrates that even a small public prototype requires systems thinking. The chassis subsystem mediates the relationship between the educational display of the system, the transportation requirements of the system, maintenance of the system, and ethical concerns associated with the use of the system.

Agile project management tools, especially Kanban and sprint-based tasking, were valuable because they allowed work to be reallocated when fabrication problems and integration delays emerged. Values thinking was most useful when design conflicts were translated into explicit criteria, particularly in balancing transparency against robustness. The LCA and FMEA further showed that seemingly local chassis decisions, such as material choice, rear access, and chute geometry, had wider implications for repairability, safety, and long-term sustainability.

References

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